

Natural Orbitals : Theory and Applications

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I Theory on Natural Orbitals

This part of the notes are mainly inspired from [1, 2, 3].

Natural orbitals correspond to a change of basis of the naïve star representation of the Anderson impurity model (AIM). This transformation allows us to separate the bath into two parts: an active part and an inactive part, the latter being considered as a single Slater determinant and therefore becoming trivial for subsequent calculations (DMRG, ED, ...). Natural orbitals thus reduce the number of degrees of freedom to those associated with the active bath sites. In the MPS framework, this reduction can significantly reduce the required bond dimension.

Let us consider a one-band, spin-independent AIM (see also Fig. 1a):

$$\hat{H}_\star = U\hat{n}_{0\uparrow}\hat{n}_{0\downarrow} + \epsilon_0(\hat{n}_{0\uparrow} + \hat{n}_{0\downarrow}) + \sum_{l=1,\sigma}^{N_{\text{bath}}} \epsilon_l \hat{n}_{l\sigma} + V_l(\hat{c}_{0\sigma}^\dagger \hat{c}_{l\sigma} + h.c.) = U\hat{n}_{0\uparrow}\hat{n}_{0\downarrow} + \hat{H}_\star^{1B} \quad (1)$$

where site 0 corresponds to the impurity and $\hat{H}_\star^{1B}$ the one-body part of the AIM. We assume that we have access to the correlation matrix $C_{ij\sigma}$ (or density matrix) of the ground state $|\psi_{GS}\rangle$ of the AIM, calculated using DMRG:

$$C_{ij} = \langle \psi_{GS} | \hat{c}_i^\dagger \hat{c}_j | \psi_{GS} \rangle \quad (2)$$

where we omit the spin index. The correlation matrix can also be written in terms of its impurity and bath components:

$$C = \begin{pmatrix} \rho_{\text{imp}} & \rho_{ib} \\ \rho_{bi} & \rho_{\text{bath}} \end{pmatrix} \quad (3)$$

The next step consists of diagonalizing the bath part of the density matrix, ρ_{bath} . We do not diagonalize the full correlation matrix in order to avoid mixing the impurity site with the bath sites and therefore losing the locality of the impurity.

Because of the local interaction on the impurity, the new occupations are not strictly equal to 0 or 1. However, most of them remain very close to these values and can therefore be considered as part of a single Slater determinant below a threshold ϵ , a tunable parameter controlling the trade-off between accuracy and computation cost. Furthermore, from the properties of Gaussian states, there exists a single eigenvalue equal to $1 - \nu_0$, with $\nu_0 = C_{00}$ being the impurity occupation [MV: still true for interacting calculations ?], which is called the *active site* $|b\rangle$. In ascending order,

$$P_{\text{bath}}^\dagger \rho_{\text{bath}} P_{\text{bath}} = \text{diag}(\nu_1, \nu_2, \dots, \nu_{N_{\text{empty}}}, \dots, \nu_{\text{active}}, \dots, \nu_{N_{\text{filled}}}, \dots, \nu_{N_{\text{bath}}}) \quad (4)$$

where we define the inactive empty bath part $\mathcal{E}_I$ as $i \in \{1, \dots, N_{\text{empty}}\}$, for which $\nu_i < \epsilon$, and the inactive filled bath part $\mathcal{F}_I$ as $j \in \{N_{\text{filled}}, \dots, N_{\text{bath}}\}$, for which $\nu_j > 1 - \epsilon$. The active part $\mathcal{A}$ is defined by $k \in \{N_{\text{empty}} + 1, \dots, N_{\text{filled}} - 1\}$. The first two spaces will eventually be treated as single Slater determinants, once they are fully decoupled from the active and impurity sites at the end of

the natural orbital transformation.

For now, we consider the (almost) empty space $\mathcal{E}$ as $m \in \{1, \dots, N_{\text{active}} - 1\}$ and the (almost) filled space $\mathcal{F}$ as $n \in \{N_{\text{active}} + 1, \dots, N_{\text{bath}}\}$. Applying this change of basis gives (Fig. 1b)

$$\hat{H}' = P^\dagger \hat{H}_\star^{1B} P, \quad P = \begin{pmatrix} 1 & 0 \\ 0 & P_{\text{bath}} \end{pmatrix} \quad (5)$$

After this transformation, the impurity and active states are still strongly hybridized with the empty and filled states. However, the hopping between the empty $\mathcal{E}$ states and the filled $\mathcal{F}$ states is now almost zero (Fig. 1b). We therefore want to preserve the separation between the empty and filled states while further reducing the hybridization between the impurity and the active site. This is necessary in order to construct an inactive space that is decoupled from the impurity and can therefore be treated trivially as a single Slater determinant. This can be achieved by performing a new unitary rotation within the subspace formed by the impurity and the active site. The goal of this rotation is to further separate $\hat{H}'$ into two *empty* and *filled* blocks. The resulting rotated states are called the *bonding* and *antibonding* states, with occupations approximately equal to 1 and 0, respectively:

$$|B\rangle = \frac{1}{\sqrt{2}} |\text{imp}\rangle + \frac{1}{\sqrt{2}} |b\rangle \quad (6)$$

$$|AB\rangle = \frac{1}{\sqrt{2}} |\text{imp}\rangle - \frac{1}{\sqrt{2}} |b\rangle \quad (7)$$

[MV: $\langle AB | \rho | B \rangle$ is not necessarily euqal to zero, right ?]

We finally reorder the sites such that the Hamiltonian takes a two-block form:

$$1, 2, \dots, N_{\text{active}} - 1, \text{antibonding}, \text{bonding}, N_{\text{active}} + 2, \dots, N_{\text{bath}} \quad (8)$$

where the two transformations (antibonding/bonding rotation and reordering) can be expressed by the following unitary matrix:

$$C = \begin{pmatrix} 0 & & & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & & & \\ 1 & 0 & & & & & & \\ & \ddots & \ddots & & & & & \\ & & 1 & 0 & & & & \\ & & & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & & & \\ & & & & & 1 & & \\ & & & & & & \ddots & \\ & & & & & & & 1 \end{pmatrix} \quad (9)$$

so that the new hamiltonian (Fig. 1c) can be expressed as

$$\hat{H}'' = C^\dagger P^\dagger \hat{H}_\star^{1B} P C = \begin{pmatrix} H_{\text{empty}} & 0 \\ 0 & H_{\text{filled}} \end{pmatrix} \quad (10)$$

Last step corresponds to tridiagonalize each block using Lanczos or Householder transformation, and then undo the transform C to recover the active and impurity site (keep locality !) (Fig. 1d)

$$\hat{H}_{NO}^{1B} = C Q^\dagger C^\dagger P^\dagger \hat{H}_\star^{1B} P C Q C^\dagger; \quad Q = \begin{pmatrix} Q_{\text{empty}} & 0 \\ 0 & Q_{\text{filled}} \end{pmatrix} \quad (11)$$

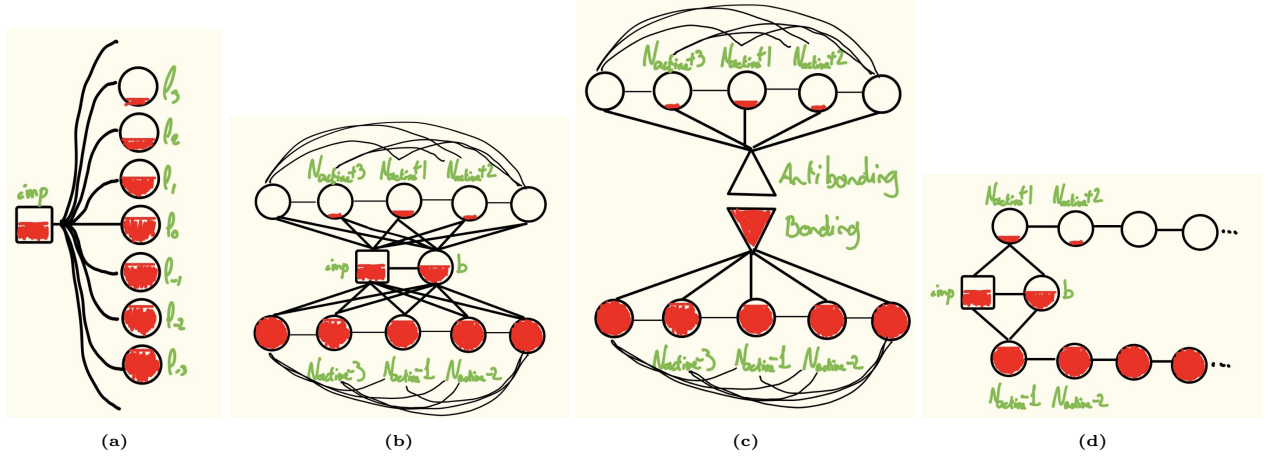


Figure 1: (a) Star representation of the AIM. (b) First rotation of $\hat{H}_*$ consisting of diagonalizing the bath density matrix. (c) Rotation of the impurity and active states to bonding and anti-bonding states with occupation of 1 and 0 respectively. (d) Hamiltonian in natural orbital basis after a tridiagonalization of the empty and filled parts and a rotation back to impurity and active states.

Therefore, the natural orbital transformation constructs two chains, one corresponding to the (almost) empty states and the other to the (almost) filled states. The impurity and active sites, which remain coupled to each other, are "located" next to the first site of each chain and are connected by a hopping term (Fig. 1d). The full natural-orbital Hamiltonian takes the form

$$\begin{aligned}
 \hat{H}_{NO} = & U\hat{n}_{0\uparrow}\hat{n}_{0\downarrow} + \epsilon_0(\hat{n}_{0\uparrow} + \hat{n}_{0\downarrow}) + \sum_{\sigma} \left(V_{0N_{\text{active}}-1}\hat{c}_{0\sigma}^{\dagger}\hat{c}_{N_{\text{active}}-1\sigma} + V_{0N_{\text{active}}+1}\hat{c}_{0\sigma}^{\dagger}\hat{c}_{N_{\text{active}}+1\sigma} \right) \\
 & + \epsilon_b(\hat{n}_{b\uparrow} + \hat{n}_{b\downarrow}) + \sum_{\sigma} \left(V_{bN_{\text{active}}-1}\hat{c}_{b\sigma}^{\dagger}\hat{c}_{N_{\text{active}}-1\sigma} + V_{bN_{\text{active}}+1}\hat{c}_{b\sigma}^{\dagger}\hat{c}_{N_{\text{active}}+1\sigma} \right) + \sum_{\sigma} V_{0b}\hat{c}_{b\sigma}^{\dagger}\hat{c}_{0\sigma} \\
 & + \sum_{m=1,\sigma}^{N_{\text{active}}-1} \left(\epsilon_m\hat{n}_{m\sigma} + t_{mm+1}\hat{c}_{m\sigma}^{\dagger}\hat{c}_{m+1\sigma} \right) + \sum_{n=N_{\text{active}}+1,\sigma}^{N_{\text{bath}}} \left(\epsilon_n\hat{n}_{n\sigma} + t_{nn+1}\hat{c}_{n\sigma}^{\dagger}\hat{c}_{n+1\sigma} \right)
 \end{aligned}$$

For clarity, the complex conjugate of each off-diagonal term is not explicitly written. For the second line, the sum over the spin corresponds to the hopping between the impurity and the first site of the empty chain, with index $N_{\text{active}} - 1$ (resp. the first site of the filled chain, with index $N_{\text{active}} + 1$). For the third line, the first term corresponds to the on-site energy of the active site b . The second term describes the hopping between the empty and filled chains, while the last term corresponds to the hopping between the active site and the impurity. Finally, for the last line, the first term corresponds to the on-site energies of the filled chain and to the hopping between nearest-neighbor sites. The same applies to the second term for the empty chain. We can therefore rewrite the one-body matrix in a more compact form:

$$\hat{H}_{NO}^{1B} = \begin{pmatrix} \epsilon_0 & & & & & & & & & \\ & \epsilon_1 & t_{12} & & & & & & & \\ & t_{12} & \ddots & & & & & & & \\ & & & \ddots & & & & & & \\ V_{0N_{\text{active}}-1} & & & \epsilon_{N_{\text{active}}-1} & V_{N_{\text{active}}-1b} & & & & & \\ V_{0b} & & & V_{N_{\text{active}}-1b} & \epsilon_b & V_{N_{\text{active}}+1b} & & & & \\ V_{0N_{\text{active}}+1} & & & & V_{N_{\text{active}}+1b} & \epsilon_{N_{\text{active}}+1} & \ddots & & & \\ & & & & & \ddots & \ddots & & & \\ & & & & & & & t_{N_{\text{bath}}-1,N_{\text{bath}}} & & \\ & & & & & & & t_{N_{\text{bath}}-1,N_{\text{bath}}} & \epsilon_{N_{\text{bath}}} & \end{pmatrix}$$

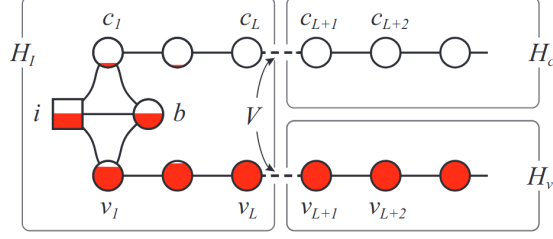


Figure 2: Separating the full Anderson impurity Hamiltonian H_A defined on the natural orbitals into four parts H_I , H_c , H_v , and V . Each Hamiltonian acts on the sites enclosed by its corresponding box. The hybridization operator V (dashed bonds) connects H_I to H_c and H_v

The computational complexity still grows exponentially with the number of bath sites. As discussed above, by introducing a threshold ϵ , we can truncate the trivial parts of the transformed Hamiltonian. These parts correspond to the operator $\hat{H}_v$ in Fig. 2 for the “inactive” filled sites, with $L+1$ analogous to N_{filled} , and to the operator $\hat{H}_c$ in Fig. 2 for the “inactive” empty sites, with $L+1$ analogous to N_{empty} . Since the occupations of these orbitals are very close to 1 or 0, respectively, they can be accurately approximated as fully occupied or empty orbitals and therefore treated as a single Slater determinant. Truncating these inactive parts gives us access to the relevant part of the Hamiltonian, namely the active part represented by $\hat{H}_I$ in Fig. 2, which contains the non-trivial many-body physics of the system. In this way, the number of bath sites that needs to be explicitly treated in the many-body calculation can be reduced to only a few sites, with the required number depending on the chosen threshold ϵ and consequently, on the desired accuracy of the calculation.

References

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